

Gowthami Jampana
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Professional Summary

Results-driven Software Engineer and Data Analyst with 2+ years of experience in Java development, data analytics, and cloud-based data engineering. Skilled in Java, SQL, Python, Power BI, and Tableau, with hands-on experience in Azure, AWS, and GCP. Proficient in ETL pipeline design, data modeling, and real-time processing using Kafka, Jenkins, and Docker. Executed academic projects in Health Analytics, Supplier Evaluation (Walmart), IoT Automation, and Digital Security, integrating data engineering and software development to deliver scalable, insight-driven solutions.

EDUCATION:

University of New Haven, West Haven, Connecticut	May 2023-Dec 2024
Master's in business analyst	CGPA: 3.5/4.0
Lakireddy Bali Reddy college of engineering, mylavaram, India	Jul 2021
Bachelor of Electrical and Electronics Engineering	CGPA: 7.2/10.0

TECHNICAL SKILLS:

Category	Skills & Tools
Core Competencies	Operations Management, Team Leadership & Development, Process Optimization, KPI Tracking & Performance Analysis, Problem-Solving & Decision-Making, Safety & Compliance, Agile & SDLC Methodologies, CI/CD Deployment
Programming Languages	C, Python, Java, PL/SQL, JavaScript,
Frameworks & Libraries	AngularJS, Node.js, React, Spring Boot, Express.js, PySpark, Pandas, NumPy
Databases	Oracle, MySQL, SQL Server, MongoDB, Redis, PostgreSQL, Snowflake
ETL / Data Engineering Tools	Azure Data Factory, AWS Glue, Apache Airflow, Apache Spark, Databricks, Hive, Hadoop (HDFS, MapReduce), Kafka
Cloud & DevOps Technologies	Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP), Git, Jenkins, Docker, Kubernetes, Terraform, Azure DevOps
Data Analytics & Visualization	Power BI, Tableau, Excel, R, Matplotlib, Seaborn,
Networking & APIs	RESTful APIs, HTTP, UI Integration, Tomcat, Flask, JSON
Testing & Automation Tools	Postman, JUnit, TestNG, TFS, Swagger UI,
Enterprise & Monitoring Tools	Apache Kafka, GitHub, Jira, Rally, Splunk

PROFESSIONAL EXPERIENCE:

Client: Tata Consultancy Services, India	July 2021 - Aug 2023
Role: Software Engineer	
Responsibilities	
Developed and maintained Java-based backend services and APIs, ensuring scalability, performance, and secure integration with front-end and data systems.	

- Executed API testing and automation scripts using JavaScript and Postman, supporting a secure transition from SAML to JWT authentication.
- Built and optimized ETL data pipelines for ingesting and transforming large datasets from APIs and databases using Java, SQL, and Python.
- Utilized MongoDB, Redis, and MySQL for data storage, caching, and real-time analytics, improving query performance and application response times.
- Deployed applications using Docker and Kubernetes, implementing CI/CD pipelines in Jenkins for continuous integration and delivery.
- Integrated Apache Kafka for real-time event streaming and message-driven data processing.
- Created comprehensive test cases and automation suites in TFS and Selenium to ensure reliability and seamless system functionality.
- Designed responsive, interactive web components using HTML, CSS, Bootstrap, AngularJS, and Media Query, ensuring cross-device compatibility.
- Collaborated with backend teams to implement RESTful APIs, troubleshoot issues, and ensure smooth front-end and back-end integration.
- Performed data analysis and visualization using Power BI and Tableau, turning raw datasets into actionable business insights.
- Monitored production systems, tracked defects using Rally, and contributed to continuous performance optimization and bug resolution.

Environment: Java, Python, MySQL, MongoDB, Redis, Apache Kafka, Azure, Docker, Kubernetes, Jenkins, Postman, Selenium, Power BI, Tableau, HTML5, CSS3, Bootstrap, jQuery, DHTML, JSON, RESTful APIs, Spring Boot (for backend integration), Media Query

ACADEMIC PROJECTS:

Master's Capstone Project – Health Risk Protection:

- Led an end-to-end data analytics project focused on assessing health risk protection and financial stability among vulnerable populations.
- Designed and implemented a data analysis framework using MySQL, building optimized queries and stored procedures for large-scale data processing.
- Extracted, cleansed, and transformed complex datasets from multiple sources to ensure data accuracy, consistency, and reliability.
- Applied statistical and predictive modeling techniques to identify key socio-economic, behavioral, and environmental risk factors.
- Developed interactive dashboards in Power BI and Tableau to visualize correlations between health indicators and financial resilience.
- Presented actionable insights to academic mentors and peers, improving understanding of risk prediction and resource allocation.

Supplier Analytics & Procurement Optimization – Walmart Project:

- Built and optimized SQL queries and stored procedures to analyze supplier performance across key dimensions such as cost, quality, reliability, and delivery timelines.
- Developed interactive Power BI and Tableau dashboards to visualize procurement metrics, contract efficiency, and vendor performance trends.
- Conducted cost-benefit and risk analyses using data models to support strategic decision-making and vendor selection.
- Created a data-driven supplier evaluation framework, integrating KPIs for financial stability, sustainability, and service levels.
- Collaborated with Walmart's procurement and analytics teams to automate supplier performance tracking and improve operational visibility.
- Utilized Python scripts for data cleansing, validation, and transformation to enhance data accuracy and consistency across ETL processes.

- Achieved measurable cost savings and processed improvements by leveraging analytics-driven insights for contract negotiations and vendor consolidation.

Smart Home Automation System Using IoT:

- Developed a smart home automation system enabling users to remotely control and monitor household appliances via a mobile application.
- Programmed IoT-enabled devices and microcontrollers using C for real-time communication between sensors, actuators, and Wi-Fi modules.
- Utilized Python for backend logic, data processing, and cloud integration to enable seamless synchronization and device control.
- Integrated motion, temperature, and light sensors with actuators to automate lighting, heating, and appliance control based on environmental data.
- Designed and implemented a user-friendly mobile interface for real-time device status, scheduling, and performance monitoring.
- Built cloud-based connectivity for secure data storage, remote access, and scalable control across multiple IoT devices.
- Enhanced energy efficiency by automating device responses to environmental conditions, reducing manual intervention and power consumption.
- Tested, debugged, and optimized embedded programs to ensure low-latency response and reliable wireless communication

DC-DC Buck-Boost Converter:

- Designed and implemented a DC–DC buck-boost converter to regulate voltage levels for electronic devices with varying power requirements.
- Developed circuitry capable of stepping up or stepping down DC voltage automatically based on dynamic load conditions.
- Ensured stable output voltage across a wide range of input levels through optimized switching and feedback control mechanisms.
- Utilized MATLAB and Proteus to simulate and analyze converter performance, validating efficiency and control response.
- Conducted detailed evaluation of ripple voltage, transient response, efficiency, and thermal behavior to enhance overall reliability.
- Optimized component selection and circuit topology to achieve high power efficiency and minimal signal distortion.
- Documented design parameters, simulation outcomes, and performance results to support reproducibility and scalability.

Security System Using XOR and NOR Gates:

- Developed a digital access control system using XOR and NOR logic gates to authenticate users based on predefined input combinations.
- Designed and implemented a cost-effective and reliable circuit that grants access only to authorized users.
- Constructed and tested the hardware prototype, validating real-time access control performance and system accuracy.
- Utilized combinational logic design principles to ensure high reliability, minimal component usage, and low power consumption.
- Created a low-cost, scalable security solution suitable for small-scale organizations, labs, and educational environments.
- Documented design schematics, truth tables, and verification results to demonstrate successful implementation and testing outcomes.

Certifications:

- NPTEL- Certificate in Cloud Computing.
- Industrial Automation Training– Indwell Automation in 2019.
- Programming for everybody certification- Coursera in 2020.
- Certificate on Joy of Computing Using Python.